Lab Work 1:

Random Line Generator

Objective

The objective of this lab is to understand the basic concepts of coordinate geometry, line representation, slope calculation, angle measurement, and graphical visualization using Python.

Learning Outcomes

After completing this lab, students will be able to:

- Generate random coordinate points.
- Create straight lines using two points.
- Calculate the slope of a line.
- Determine the angle a line makes with the positive x-axis.
- Visualize multiple lines on a graph.
- Label points and interpret graphical outputs.
- Use Python libraries such as NumPy and Matplotlib for mathematical visualization.

Problem Statement

Develop a Python program that generates **N random straight lines**. For each line:

1. Randomly generate two distinct points.
2. Draw the line connecting those points.
3. Calculate the slope of the line.
4. Calculate the angle between the line and the positive x-axis.
5. Display the line information in the console.
6. Plot all generated lines on a graph using different colors.
7. Label all start and end coordinates on the graph.

Requirements

Input

- Number of lines (**N**) to generate.

Processing

For each line:

- Generate two unique random points.
- Ensure the start point and end point are not identical.
- Calculate:
  - Slope
  - Angle with the positive x-axis
- Store the information for visualization.

Output

Display:

- Start Point
- End Point
- Slope
- Angle with x-axis
- Count positive and negative slope lines.
- Identify the steepest line.
- Calculate the average slope.

Generate a graph showing:

- All lines
- Coordinate labels
- Grid
- Axis labels
- Title
- Save the graph as an image.

Mathematical Concepts

Slope of a Line : The slope represents the steepness of a line.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Where:

- (x1, y1) = Start point
- (x2, y2) = End point

Angle with Positive X-Axis

The angle can be calculated from the slope using:

$$\theta = \tan^{-1}(m)$$

Convert the result from radians to degrees.

Suggested Algorithm

1. Read the value of **N**.
2. Repeat N times:
   - Generate two random points.
   - Verify that the points are different.
   - Calculate slope.
   - Calculate angle.
   - Store line information.
3. Print all line details.
4. Plot all lines on a graph.
5. Label coordinates.
6. Display the graph.

Graph Requirements

The graph should contain:

- Different color for each line
- Coordinate labels for start and end points
- Grid lines
- X-axis label
- Y-axis label
- Graph title

Sample Output Format

Line 1
Start Point : (2,4)
End Point : (7,9)
Slope : 1.00
Angle : 45.00°

Line 2
Start Point : (1,8)
End Point : (5,2)
Slope : -1.50
Angle : -56.31°

Total count of positive and negative slope lines: 3 & 1
The steepest line: Line 2
Average slope: 1.4

Submission Requirements

Students must submit:

1. Source code
2. Screenshot of program execution
3. Screenshot of generated graph
4. Short report containing:
    - Objective
    - Algorithm
    - Sample Input/Output
    - Graph Output
    - Conclusion

Deadline: As announced in class.